

Sketch-Note Guide: Valuation & Analysis of Bonds with Embedded Options

A visual, formula-rich summary for CFA Level II.
Focus: Callable & Putable Bonds, Risk, and Volatility.

THE BUILDING BLOCKS (VALUATION RELATIONSHIPS)

STRAIGHT (OPTION-FREE) BOND



The baseline.
No contingency provisions.
Certain cash flows.
Foundation for valuation.

CALLABLE BOND VALUE

$$\text{VALUE} = \text{STRAIGHT BOND VALUE} - \text{CALL OPTION VALUE}$$



CALL
Investor is 'short' the call.
Issuer holds option.
Call value reduces bond price.

PUTABLE BOND VALUE

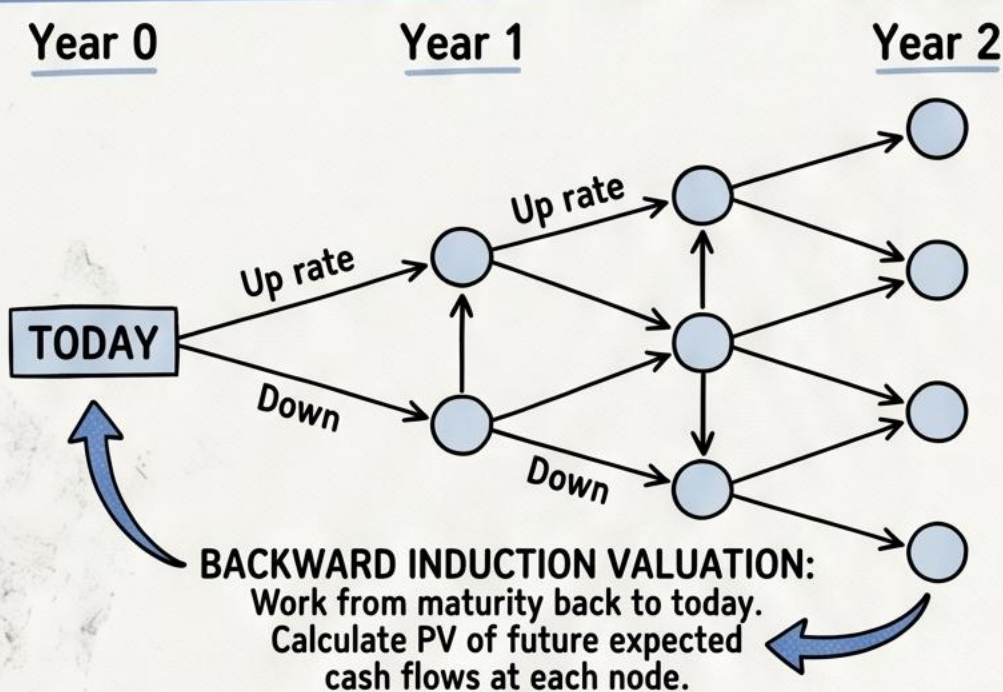
$$\text{VALUE} = \text{STRAIGHT BOND VALUE} + \text{PUT OPTION VALUE}$$



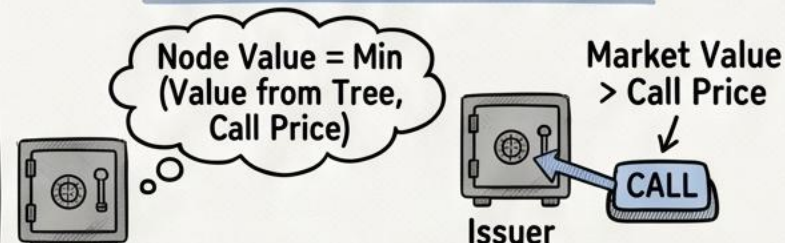
PUT
Investor is 'long' the put.
Option increases bond value.

★ Requires arbitrage-free framework due to uncertain cash flows dependent on rates.

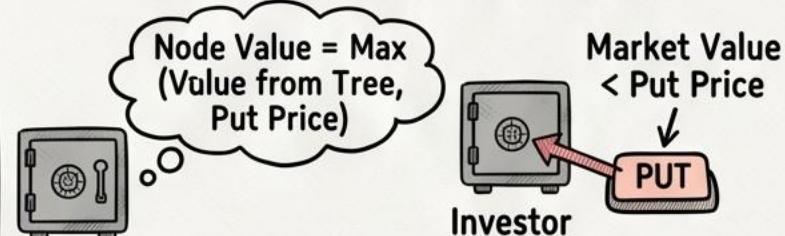
BINOMIAL TREE MECHANICS & NODE DECISIONS



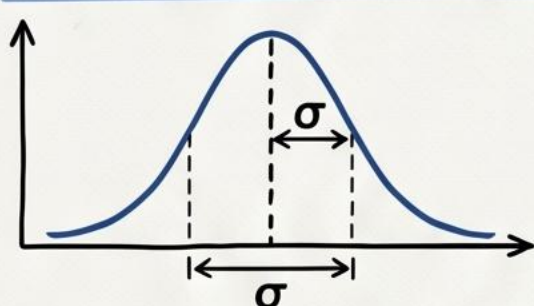
THE 'CALL' NODE DECISION



THE 'PUT' NODE DECISION



THE VOLATILITY FACTOR (σ)



Volatility
UP ($\sigma \uparrow$)

OPTION VALUE UP

CALL $\uparrow$ PUT $\uparrow$
More opportunities to be 'in the money'.

BOND PRICE IMPACT

$\downarrow$ CALLABLE PRICE DOWN (Straight - Call Option)
 $\uparrow$ PUTABLE PRICE UP (Straight + Put Option)

RISK METRICS (DURATION & CONVEXITY)

EFFECTIVE DURATION (EffDur)

$$\text{\$EffDur} = \frac{[PV_- - PV_+]}{2 \times \Delta \text{Curve} \times PV_0}$$

The only appropriate measure!
Accounts for changing cash flows.

★ Callable = High Up-Duration. Putable = High Down-Duration (Asymmetrical).

EFFECTIVE CONVEXITY (EffCon)

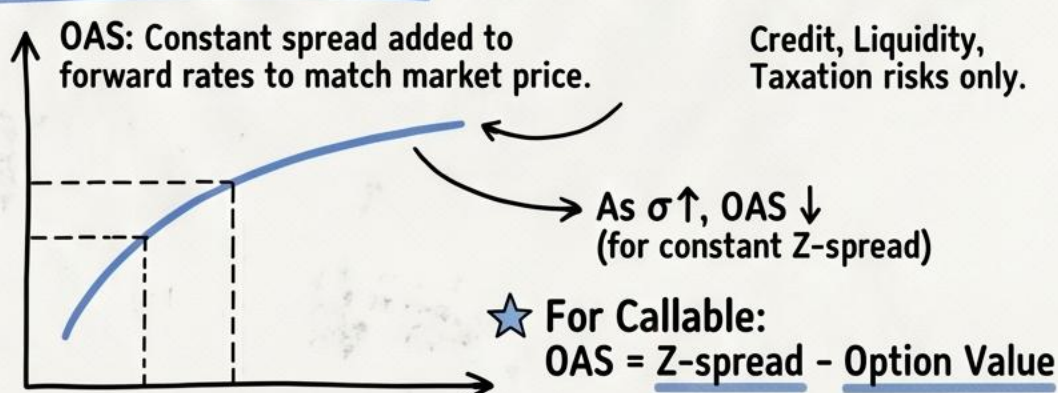
$$\text{\$EffCon} = \frac{[PV_- + PV_+ - (2 \times PV_0)]}{[(\Delta \text{Curve})^2 \times PV_0]}$$

Callable: NEGATIVE CONVEXITY at low yields (capped upside).
Putable: High Positive Convexity.

EFFECTIVE DURATION COMPARISONS

	BOND TYPE	CHARACTERISTIC
1	Straight Bond	Highest; stable across shifts.
2	Callable Bond	$\leq$ Straight; drops significantly when rates fall (near call price).
3	Putable Bond	$\leq$ Straight; drops significantly when rates rise (near put price).

OAS & SPREADS (THE 'SPREAD' LOGIC)



EXAM TRAPS & PRO-TIPS

- ✓ THE 'BUSTED' CONVERTIBLE:
Stock Price $\ll$ Conversion Price $\rightarrow$ Bond-like behavior.
Option is for 'out of the money'.
- ✓ Z-SPREAD VS. OAS:
OAS < Z-spread for Callable.
If OAS > Z-spread $\rightarrow$ Putable (Option value adds to spread).
- ✓ KEY RATE DURATION:
Total Duration = $\sum$ Key Rate Durations.
Useful for non-parallel shifts.

SKETCH & LEARN! CFA L2